

1. Which are the advantages of using IDSs at the network level?

- a) Ensures the protection of electronic transactions, detecting unauthorized change to data
- b) Does not affect the performance of the supervised systems
- c) Can monitor any traffic, including IPsec and SSL
- d) High scalability (with a few well-placed sensors it is possible to monitor an entire network)

2. Which tool do you use when you need to detect the running services on a computer?

- a) Port scanner
- b) IDS at the host level
- c) Packet sniffing
- d) File integrity checker

3. Which of the following statements about signature-based IDSs are true?

- a) Generating a relatively low number of false positive alarms
- b) Provides detailed info about the type of attack
- c) Requires trained personnel to investigate the alarm
- d) Can detect new types of attack

4. Which application-level protocol uses UDP for data transfer?

- a) DNS
- b) LDAP
- c) HTTP
- d) SNMP

5. Which of the following products can be used to scan the existing vulnerabilities on a computer system?

- a) Snort
- b) Ethereal/Wireshark
- c) Nessus
- d) Nmap

6. What protocol is used to discover the MAC address for an IP address?

- a) ARP
- b) ICMP
- c) DNS
- d) DHCP

7. What IP class is the IP address 190.221.78.10?

- a) Class A
- b) Class C
- c) Class D
- d) Class B

8. Which are not routable IP addresses on the Internet?

- a) 192.169.30.20
- b) 192.128.0.10
- c) 120.15.250.5
- d) 172.31.250.24

9. What is happening in a TCP handshake?

- a) The sender computer sends a TCP packet with SYN and ACK flag set
- b) The sender computer sends a TCP packet with SYN flag set
- c) The receiver computer sends a TCP packet with ACK flag set
- d) The receiver computer sends a TCP packet with SYN flag set

10. What is a firewall?

- a) A complete security architecture
- b) A control and monitoring point for the network traffic
- c) Best single solution for all network's security problems
- d) Practical implementation of network security policy

11. When a frame is received by a router network interface, what address is changed inside the frame?

- a) IP address
- b) MAC address
- c) Port address
- d) IPX address

12. DMZ is:

- a) Intranet network segment protected by a firewall, where the private services are available
- b) A network segment between Internet and the intranet
- c) A network segment where public services and application gateways are deployed
- d) A network segment in front of a firewall, which is not protected by the firewall

13. A company has a Class C and needs 5 sub-networks, each with a minimum of 20 workstations. Which subnet mask is needed?

- a) 255.255.255.240
- b) 255.255.255.0
- c) 255.255.255.92
- d) 255.255.255.224

14. What types of firewalls maintain state information for packet analysis?

- a) Stateful inspection firewall
- b) Packet filter firewall
- c) Application level gateway
- d) Circuit level gateway

15. What are the differences between IPS and IDS?

- a) Operates inline
- b) May analyze encrypted traffic
- c) Detects network viruses
- d) Is implemented only in hardware

16. The network switch is better than the hub because:

- a) Provides the minimum security...
- b) It has more network ports
- c) The traffic for a certain workstation is sent to all workstations
- d) It ensures a larger bandwidth per port

17. Which fields from the IP headers are used for infinite loop prevention within the network packets?

- a) time-to-live
- b) header checksum
- c) service type
- d) fragmentation offset

1. The Message Integrity Code (MIC):

- a) uses the same frame format as the WEP protocol
- b) uses a frame with sequence numbers
- c) the source and destination MAC addresses are not included in the MIC calculation
- d) uses only the frame's data for the calculation

2. The 802.11i protocol:

- a) does not provide a robust security service
- b) uses only the AES protocol for encryption
- c) does not support mutual authentication
- d) uses the same keys regardless of transmission time

3. Key management in the 802.11i standard is performed:

- a) only through a pairwise key
- b) during the 4-way handshake
- c) only through a pre-shared key
- d) only through a group key

4. The CCM (CCMP) protocol:

- a) uses the RC4 algorithm
- b) uses AES
- c) encrypts the entire CCMP frame
- d) uses a header composed only of the current packet number

5. SIM authentication:

- a) uses SIM cards for the RADIUS server
- b) does not use a shared key known by both the SIM and the GSM authentication center
- c) the authentication and encryption algorithms are stored on the RADIUS server
- d) ensures mutual authentication between the client and an AAA server

6. The LEAP authentication protocol:

- a) performs mutual authentication
- b) uses PKI certificates
- c) is a non-proprietary protocol
- d) uses SIM modules for authentication

Answer: None

7. MAC authentication:

- a) assumes the client verifies the access point's MAC address
- b) assumes the access point verifies the client's MAC address
- c) assumes both verifications
- d) assumes using an EAP protocol to verify MAC addresses

8. In authentication using the EAP protocol, the uncontrolled port is:

- a) a port that allows only the transmission of authentication messages
- b) a port that allows the transmission of other data as well
- c) a port opened only after the work session ends
- d) a port that is used only for data representing MAC addresses

9. Protection against a denial-of-service (DoS) attack is achieved by:

- a) requiring end-to-end mutual authentication
- b) aiming directional antennas outward from the building
- c) deliberately occupying the transmission channels
- d) using the victim node as an intermediary to continuously retransmit useless data packets

10. Stand-alone access points:

- a) have centralized management for transmit power and frequencies
- b) have high acquisition, installation, and operational costs
- c) can operate independently, providing authentication, security, and management in a single unit
- d) are not recommended for small and medium businesses

11. The regeneration of encryption keys at specified intervals:

- a) ensures error-free transmission
- b) is a methodology that does not require RADIUS servers
- c) ensures simpler management of network security
- d) ensures increased data confidentiality

12. Grouping into work domains of wireless networks:

- a) assumes creating cells only at the department level
- b) assumes a wireless service using the same SSID
- c) assumes creating cells only with the same type of access point
- d) assumes creating cells only with the same type of client

13. The RADIUS authentication system:

- a) uses only servers with centralized databases
- b) uses only servers with distributed databases
- c) uses only servers with databases linked together
- d) uses servers with distributed or centralized databases

14. Data link layer traffic monitoring devices can be countered by:

- a) encrypting the transmitted data
- b) analyzing the traffic flow
- c) using powerful servers
- d) using non-overlapping channels

15. Protection against rogue (false) access point attacks:

- a) is achieved by using high operating speeds
- b) is achieved by using mutual authentication
- c) is achieved by using multi-system access points
- d) is achieved by using cells by domain

16. In the case of wireless computer networks, we can have:

- a) passive attacks
- b) active attacks
- c) passive and active attacks
- d) topological attacks

17. A session hijacking attack involves:

- a) altering packets from the access point to the client
- b) copying the access point's MAC address, de-authenticating the client, and relaying data between the attacker and tc)
- c) disabling the access point's operation
- d) establishing the link between the attacker and the client

18. A man-in-the-middle attack involves:

- a) the attacker becoming an access point for the client and a client for the real access point
- b) the attacker changing the SSID of the access point
- c) creating a new cell
- d) extending the existing cell

19. The WEP protocol:

- a) uses a different algorithm for the sent message and the received message

- b) does not have a field to protect against unauthorized modifications
- c) encrypts all fields of the frame
- d) does not encrypt the IV field

20. The SSID identifier:

- a) is an identifier only for the access point
- b) is an identifier only for the client
- c) is an encrypted message in the Beacon frames
- d) is an identifier that provides the ability to logically separate wireless networks

21. Shared-key authentication:

- a) presumes authentication of the client
- b) presumes authentication of the access point
- c) presumes authentication of both the access point and the client
- d) presumes that the access point has an authentication database

22. The Initialization Vector (IV) in the WEP protocol:

- a) this field is used to calculate the ICV field
- b) it is a field that is not added to the static key, being used autonomously
- c) it is an optional field
- d) it is a field that is added to the static key

23. The 802.1X authentication standard:

- a) is a port-based standard with an authentication server
- b) is a standard based only on ports
- c) is a standard based only on the authentication server
- d) is a standard based on device authentication

24. The EAP protocol uses:

- a) only lower-level EAPOL packets
- b) only application-level RADIUS packets
- c) both types of packets
- d) is used only by wireless networks

25. The EAP-MD5 protocol:

- a) is a certificate-authority-based protocol
- b) protects the message exchange by creating a unique fingerprint that is digitally signed
- c) uses PKI certification

d) allows elimination of man-in-the-middle attacks

Answer: None

26. The Extended Authentication Protocol with Transport Layer Security (EAP-TLS):

- a) is a protocol based on PKI certificates, requiring both the server and the client to present certificates
- b) does not perform mutual authentication
- c) does not rely on X.509 certificates
- d) is based on username and password authentication

27. The Protected Extensible Authentication Protocol (PEAP):

- a) ensures authentication based on username and password
- b) does not support mutual authentication
- c) the PEAP client authenticates directly to the authentication server
- d) does not support fragmentation

28. The WPA protocol:

- a) uses WEP-based encryption
- b) provides confidentiality using the TKIP protocol
- c) ensures message integrity using the ICV
- d) uses an initialization vector like WEP

29. The Temporal Key Integrity Protocol (TKIP):

- a) uses single-phase key mixing
- b) does not use a sequence counter
- c) uses the MIC for message integrity
- d) encrypts the entire MPDU frame, including the header